What remains to formalize in Lean
A section-by-section assessment of the $c = 8$ draft

Companion assessment based on the local Lean and Mathlib sources

27 September 2026

Abstract

The existing Lean proofs verify a few finite algebraic implications; they do not formalize the lattice construction or the Haagerup CFT theorem. This assessment distinguishes a useful conditional lemma from an end-to-end formalization, records reusable library infrastructure, and assigns conversion difficulty to every section of the manuscript. The original coverage assessment is retained, with navigation updated to the current manuscript's sections and appendices; an additional assessment covers the physics sections and their new exact datasets. The frozen original remains available with the archived baseline. Two additional Lean lemmas check the real-number rearrangement and positive division used in the thermal tail bounds, with their analytic premises explicit. The Python arithmetic certificates and outward-rounded intervals remain separate from these kernel-checked implications. The estimates describe dependency and proof-engineering difficulty, not a prediction of elapsed time or a claim that no relevant external formalization exists.

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Version and manuscript links. All operative section, appendix, proposition, and page references below link to the current `hi_cft_c8_proof.pdf` through its stable LaTeX labels. Keep the companion and manuscript PDFs in the same directory to follow these links. The mathematical construction is now in Appendices A–L; the ledger, formalization, and build records are in Appendices M–O. The concordance retains earlier numbers only as historical identifiers for the 55-page manuscript archived at `c8_physics_20260927/baseline/hi_cft_c8_proof.pdf`. The frozen original assessment remains in `c8_reviewability_20260927/lean_assessment.tex`. Updating navigation does not change the assumptions or claimed scope of any Lean declaration.

1 Lean formalization: present coverage and remaining work

The present Lean development does not formalize the central-charge-eight construction or its main theorem. It checks several useful implications inside the finite-closure and top-bound arguments. A separate, much larger development proves existence of coefficients satisfying the Evans–Gannon equations; its categorical reconstruction and its connection to this VOA are not formalized. These are substantive results with different scopes. Neither should be described as a formal verification of the entire CFT manuscript.

This assessment reads the revised manuscript, the local Lean sources, their saved check reports, and the installed Mathlib sources. The pinned toolchain is Lean `v4.33.1`; the Mathlib revision in `lean/lakefile.toml` is `0df444a360eaa60ab8c11dca51a86af692955474`. Paths below are relative to the repository; paths beginning `Mathlib/` are relative to `lean/.lake/packages/mathlib/`. Declaration names were checked against that checkout. This is a source and scope audit, not a new compiler run or an audit of every transitive dependency.

1.1 What is already checked

All five declarations below are in the namespace `HaagerupIzumi.CFT`. The two source files are independent of the Evans–Gannon umbrella import `HaagerupIzumi.lean`.

Declaration and source	Exact scope of the conclusion
`nilpotent_raising_saturation`, `CFT/FiniteClosure.lean`	A submodule stable under two linear operators is the whole module, assuming a spanning equation modulo their two images and vanishing of every operator word of one fixed length. Joint nilpotence is an input; the theorem does not derive it from the actual current weights.
`projected_residual_recovery`, same file	From $v - Pv \in C$ and $Pv \in C$, deduce $v \in C$. No idempotence of P is needed. The actual projector coefficients and the two membership assertions are not checked by this theorem.
`jacobi_bottom_witness`, same file	From $f(v) = g(u)$, $u = 0$, $v = cq$, and $c \neq 0$, deduce $f(q) = 0$. The Jacobi identity, the relevant quotient, and the particular bottom coefficient are explicit inputs, not reconstructed VOA facts.
`core_top_deficit_identity`, `CFT/TopBound.lean`	The printed fourteen-variable polynomial identity, with the exact dimension and integer-scaled trace coefficient definitions.

Declaration and source	Exact scope of the conclusion
`core_top_bound`, same file	Nonnegative multiplicities and the two stated current-character inequalities imply the cleared-denominator 17/10 bound. It does not derive the fourteen tops, their multiplicities, or their stress traces.

The full file paths have prefix `lean/HaagerupIzumi/`. The reports `c8_revision_20260927/lean_finite_closure_final.log` and `c8_revision_20260927/lean_top_bound.log` record successful checks. Their axiom reports contain only `propext`, `Quot.sound`, and, where used, `Classical.choice`. The projected-residual lemma has an empty axiom report. These are the usual logical foundations of the stated Lean proofs, rather than new CFT-specific assumptions. The rational reconstruction of the top table in `c8_revision_20260927/check_top_bound.py` is separate from the Lean proof of the inequality.

1.2 Assessment of every manuscript section

The difficulty descriptions distinguish a *local conditional formalization*, in which the surrounding mathematical structures and theorems are supplied explicitly, from an *end-to-end formalization*, which constructs those structures and proves the inputs. "Focused" means a bounded algebraic or analytic lemma with nearby library support. "Substantial" means several interacting new definitions and proofs. "Foundational" means that major mathematical theories or deep external theorems still have to be formalized. These are assessments of dependency scope, not estimates of working hours.

Current anchor	Existing CFT coverage	Missing work and conversion difficulty
Section 1 p. 4	No formal main theorem or definition of the constructed U.	A theorem assembling explicit interfaces would be focused once its inputs exist. A proof of the stated existence theorem inherits all the foundational dependencies below; packaging them as assumptions does not discharge them.
Appendix A p. 28	No lattice VOA, cocycle-to-field construction, or embedding is formalized here.	Finite lattice arithmetic, the four-component coefficient algebra, and low-weight identities are substantial but concrete targets. Construction of the Fock space, lattice fields, PCT, and their VOA identities is foundational.
Appendix B p. 31	No filtered Wick theorem or all-weight recursion is formalized.	A graded mode-algebra interface could support the induction, filtration, skew-symmetry, and Wick reductions. This is a substantial local project; deriving that interface for the actual lattice VOA adds foundational work.
Appendix C p. 34	Raising saturation and projected-residual recovery are checked as abstract implications.	Prove bounded-current-weight joint nilpotence, the singlet projection identities, and the actual rank and residual certificates. These are focused to substantial once the finite spaces and current actions are represented faithfully.

Current anchor	Existing CFT coverage	Missing work and conversion difficulty
Appendix D p. 36	Only the abstract bottom-witness scalar cancellation is checked.	Derive the quotient Jacobi relation from mode identities, justify the charged spanning and neutral-family reductions, and verify all nonzero coefficients. The local proof is substantial; most scalar identities are focused.
Appendix E p. 38	No theorem applies the gates to the actual subVOA.	Given a formal recursive criterion and genuine gate inputs, the induction, dimension comparison, and properness deduction are focused. Without the bridge from finite data to actual modes, they remain conditional.
Appendix F p. 39	No embedded Potts/parafermion VOA or regularity theorem is formalized.	Finite positive-operator inequalities and the displayed stress identity are accessible local targets. Their application requires conformal vectors, C_2 spaces, regular modules, cosets, and the relevant rationality theorems: foundational work.
Appendix G p. 44	The final fourteen-variable top-bound identity and inequality are checked.	Derive the core-top table and current inequalities; formalize branching, the rank-four exclusion, rank-six based-ring argument, formal codegrees, and center data. Conditional finite arithmetic is focused to substantial. VOA tensor theory and fusion-category classification remain foundational. The optional classification route must retain its extra hypothesis.
Appendix H p. 58	No theorem identifies the full character vector.	Finite twist arithmetic and the equation $6 = 6 + 13\gamma$ are focused. The proof needs trace characters, Zhu modularity, the specified modular representation, finite image, the principal-part uniqueness argument, and the Evans–Gannon fundamental matrix. Their complete formalization is foundational, despite existing scalar modular-form infrastructure.
Appendix I p. 59	None of the analytic comparison propositions is formalized.	Common-core reduction, dense-domain transport, analytic continuation, and the Haag-duality squeeze are bounded local targets. Conformal nets, DHR sectors, reconstruction, annulus integration, sewing, and fusion positivity constitute the principal foundational burden. Detailed distinctions appear below.
Appendix J p. 67	No unitary ribbon identification is checked.	The ring-level scalar arguments and cyclic-group obstruction calculations can be isolated. A proof of the categorical conclusion needs unitary fusion categories, the relevant Izumi classification, Morita equivalence, and the center/ribbon comparison. General rigid and braided categories in Mathlib supply only part of this framework.

Current anchor	Existing CFT coverage	Missing work and conversion difficulty
Appendix K p. 69	No Q-system extension, diagonal full theory, or E_8 identification is formalized.	Once extension correspondences and categorical duality are inputs, transport of the chosen algebra and finite dimension checks are substantial local targets. Proving the correspondences, the full-net construction, and the holomorphic-VOA classification is foundational.
Appendix L p. 71	No formal identification with another physical construction.	Most assertions are scope statements, not independent formal theorems. One can formalize the conditional character and extension comparisons after the earlier interfaces exist. Neither those interfaces nor current Lean files imply a spin-chain continuum limit or uniqueness of all $c = 8$ theories.
Appendix M p. 71	No Lean authentication or mathematical replay of the frozen data.	A small sound certificate checker with explicit data is a concrete, substantial target. Its soundness must connect encoded vectors and matrices to the actual fields. Hash agreement binds files but proves neither their residual identities nor their VOA interpretation.
Appendix N p. 74	The five construction declarations listed above, plus elementary support lemmas in their two files. The current appendix also records the two thermal lemmas assessed in the physics supplement below.	Keep theorem names, assumptions, source hashes, toolchain, and axiom reports synchronized. This is focused maintenance; the appendix must not enlarge the scope of the checked declarations.
Appendix O p. 75	Successful TeX compilation is not a Lean result.	A reproducible build manifest can record the checked artifacts and interfaces. Mathematical review, compiler checking, finite replay, and PDF compilation are separate checks with separate conclusions.

1.3 Finite certificates and the algebraic sections

The new companion now supplies concrete data for this next stage: a 120-row short-generation graph with normalization and target expansions, 33 projected Gate 4 vectors and all twelve relations, the explicit Potts and parafermion membership expressions, and the eighty-current space's matrices and cyclic-basis recipes. These are in `c8_reviewability_20260927/`, with separate successful Python coordinate, residual, and matrix checks. They remove the need to extract those witnesses again. They introduce no Lean proofs: formal parsers, checker soundness, and the interpretation of the supplied mode recipes and matrices remain to be established.

For Appendices A–E and the ledger in Appendix M, the most valuable next boundary is between the sparse finite data and the mode calculations that they purport to represent. A certificate

checker should have a proved interpretation map from its encoded vectors into a mathematical vector space. It should prove that its addition, scalar multiplication, mode actions, and selected column combinations have the stated meanings. Only then does a zero residual certify the required identity in that space. The further identification of this space and action with the actual lattice fields is a distinct theorem.

The coefficient algebra uses $1, s, t, st$ with $s^2 = 13$ and $t^2 = -3$. One can first verify multiplication and residual identities in this four-dimensional rational algebra. Rank arguments additionally need a field, or an appropriately injective embedding into the intended complex coefficient field; merely writing a quotient by the two polynomials does not establish those properties. Sparse vectors fit `Finsupp.linearCombination` in `Mathlib/LinearAlgebra/Finsupp/LinearCombination.lean`. The declarations `Matrix.rank_of_det_ne_zero`, `Matrix.rank_submatrix_le`, and `Matrix.rank_eq_finrank_span_cols` in `Mathlib/LinearAlgebra/Matrix/Rank.lean` provide useful rank interfaces. A nonzero minor together with an independently proved upper bound gives an economical exact rank certificate. A modular nonzero minor may also prove a characteristic-zero lower bound once the reduction map and cleared denominators are justified; a modular zero residual is insufficient for characteristic-zero containment.

The current-algebra part has more reusable support than a new representation-theory library would suggest. In `Mathlib/Algebra/Lie/Sl2.lean`, `IsSl2Triple.HasPrimitiveVectorWith` packages a highest-weight vector. Its lemmas `lie_h_pow_toEnd_f`, `lie_e_pow_succ_toEnd_f`, and `pow_toEnd_f_eq_zero_of_eq_nat` give the standard weight, raising-coefficient, and finite-dimensional nilpotence calculations. These can support the singlet polynomials and the bounded-weight argument supplying `nilpotent_raising_saturation`. They do not identify the present current actions with integrable affine-VOA modules. Nor does the generic saturation theorem verify the saved spanning matrix.

Mathlib already has formal vertex *operators*: `VertexOperator`, `VertexOperator.ncoeff`, and `VertexOperator.of_coeff` are in `Mathlib/Algebra/Vertex/VertexOperator.lean`. They use Hahn-series linear maps from `Mathlib/Algebra/Vertex/HVertexOperator.lean`, which provides coefficient and composition operations. This is relevant groundwork for mode indexing and lower truncation. The inspected file does not provide a complete conformal vertex algebra with vacuum, translation, Jacobi/locality, modules, and invariant forms; its own TODO list includes locality and residue products. Directory and declaration searches, followed by inspection of these files, located no existing implementation of the manuscript's complete VOA interface. That is a statement about the audited checkout, not a claim that Mathlib contains no relevant vertex-algebra ingredients.

The filtered Wick criterion is therefore a worthwhile intermediate deliverable. Its hypotheses should explicitly state the mode identities and filtration controls used in the proof. Formalizing only an abstract implication is useful, provided the final theorem continues to display those hypotheses until the actual lattice construction supplies them.

For Appendix F, the positive stress identity separates similarly. Finite-grade self-adjoint operators and their nonnegative quadratic forms can be treated with the inner-product-space library. For example, `LinearMap.IsSymmetric.eigenvectorBasis` in `Mathlib/Analysis/InnerProductSpace/Spectrum.lean` supports finite-dimensional diagonalization. Turning the inequality into regularity requires the specified conformal embeddings and the module-theoretic finiteness results. A formal proof of a positive matrix identity alone cannot establish those embeddings or C_2-cofiniteness.

Appendix G has three distinct layers. The first is the core-top enumeration and exact trace inequality; only the last linear implication is currently checked. The second is finite algebra: branching multiplicities, a based ring on four or six symbols, dimension equations, and root-of-unity sums. These are credible conditional targets using finite types, matrices, polynomial identities, and number theory. The third is the interpretation through semisimple tensor categories, formal

codegrees, indicators, induction, and a modular center. It needs structures and theorems beyond a table of fusion coefficients. In particular, verifying a fusion table does not verify a pentagon, construct its associator, or classify its categorifications.

The optional algebraic classification Proposition G.13 (p. 57) must have an explicit parameter expressing exhaustion by the four categories in the cited list. Its pivotal-scalar exclusion can then be formalized separately. The repository's Evans–Gannon coefficient construction does not discharge that exhaustion hypothesis. The manuscript's unitary route avoids this optional assumption, but a Lean version of that route still needs the separate unitary classification and Morita-equivalence theorems. Avoiding one external premise does not automatically formalize the replacement theorem.

For Appendix H, the installed library has `ModularForm` in `Mathlib/NumberTheory/ModularForms/Basic.lean` and `UpperHalfPlane.qExpansion`, `UpperHalfPlane.qExpansion_eq_zero_iff`, and `UpperHalfPlane.qExpansion_coeff_unique` in `Mathlib/NumberTheory/ModularForms/QExpansion.lean`. There are also scalar theta, eta, congruence-subgroup, and level-one developments. They do not directly supply Zhu's vector-valued characters or the exact finite-image representation used here. An appropriate first theorem would prove principal-part uniqueness under explicit transformation, holomorphy, cusp, and invariant-vector hypotheses, and then distinguish its application to the actual trace functions from finite coefficient arithmetic.

1.4 The analytic comparison in detail: A2, A3, and A1

The target is Appendix I, *From vacuum unitarity to complete unitarity*, beginning on p. 59.

The manuscript now gives mathematical proofs of all three comparison steps. Their lack of a Lean formalization is not an assertion that they remain mathematical conjectures. Conversely, replacing each of them by a Lean proposition whose proof is an assumed field would only record the dependency graph.

A2: the actual subnet and sector count. Proposition I.1 (p. 60) is the precise target. A local formalization can separate the branching multiplicities $1, 1, 2$, the endomorphism algebra $\mathbb{C}\oplus\mathbb{C}\oplus M_2(\mathbb{C})$, and the rank-six based-ring argument. Those are finite algebra once the representation objects and their decompositions are supplied. The passage from an algebraic VOA summand to a reducing Hilbert subspace requires the A3 common-core lemma, energy bounds, and an actual represented subnet. The proof of local irreducibility uses locality, strong additivity, the shared rotation action, and the separating vacuum; all four inputs must be visible in a formal statement.

End-to-end A2 then needs finite-index local extensions, DHR sectors, statistical dimension, the canonical algebra, center induction, and complete rationality. The index-two exclusion also needs the precise net-automorphism/VOA-automorphism correspondence and the finite-order Lie-algebra fixed-rank argument. Its untwisted $\mathbb{Z}/3$ subcategory comes from the transverse diagonal and antidiagonal isotropic lines, followed by free-module induction. A formal proof should preserve that construction: an abstract group of three invertible symbols does not determine a trivial associator.

Mathlib does provide `VonNeumannAlgebra` and `VonNeumannAlgebra.commutant` in `Mathlib/Analysis/VonNeumannAlgebra/Basic.lean`. Here a concrete von Neumann algebra is packaged by its bicommutant property. This is useful support for locality and commutant arguments. The file is not a conformal-net, DHR, subfactor-index, or Connes-fusion library. Those theories were not located in the audited checkout. The distinction matters: the basic operator-algebra vocabulary exists, while the net-theoretic theorems used by A2 still need formalization.

A3: reconstruction, common cores, and exhaustion. The reconstruction statement is Proposition I.3 (p. 62), with its strong-unitarity consequence in Proposition I.4. The most accessible

analytic target is the graph-reduction Lemma I.2 (p. 62). Represent a densely defined operator by `LinearPMap`, require closability, and assume explicitly that the bounded orthogonal projection preserves its domain and commutes there. The map $(\xi, \eta) \mapsto (P\xi, P\eta)$ preserves the graph and therefore its closure. This proves commutation on the closed domain. The adjoint claim follows from the defining inner-product identity, using $P^* = P$ and domain invariance. Both arguments can be formalized without claiming that the original core is a core for the adjoint.

The existing declarations `LinearPMap.IsClosable.graph_closure_eq_closure_graph`, `LinearPMap.IsClosable.closure_isClosed`, and `LinearPMap.HasCore` are in `Mathlib/Topology/Algebra/Module/LinearPMap.lean`. The complementary file `Mathlib/Analysis/InnerProductSpace/LinearPMap.lean` provides `LinearPMap.IsFormalAdjoint` and `LinearPMap.adjoint`. Projection APIs include `Submodule.orthogonalProjectionOnto` and `Submodule.starProjection` in `Mathlib/Analysis/InnerProductSpace/Projection/Basic.lean`. These are actual reusable declarations, not proposed abstractions. The further conclusion about the von Neumann algebra generated by an unbounded field requires an affiliation or polar/spectral-generation interface in addition to the graph lemma.

The next target is the energy-domain argument. Starting with a positive energy representation whose bounded-energy vectors form a weak module, regularity gives an ordinary simple summand S. One must prove that its Hilbert closure reduces L_0 and that its projection sends every bounded-energy vector into a finite sum of finite-dimensional grades of S. This establishes $PM \subset S \subset M$ before any field commutation is used. Assuming finite multiplicities for the original net representation would remove precisely the issue the proof resolves. Likewise, a smeared field need not map the bounded-energy core to itself; its value is a Hilbert-space vector. A Lean signature that accidentally makes every smeared field an endomorphism of that core would encode a stronger and inappropriate premise.

The grade-by-grade comparison of two positive invariant forms is another bounded target. It first constructs an algebraic module endomorphism, uses an eigenvector on a finite-dimensional lowest grade and simplicity to prove that endomorphism scalar, and only then obtains boundedness of the module isomorphism. This order avoids importing a Hilbert-space Schur lemma before boundedness is known. Finite direct sums and the counting argument that twelve reconstructed simple modules exhaust twelve algebraic simples are comparatively small after the interfaces are fixed.

Full A3 still requires the Henriques–Tener reconstruction theorem, including its bounded-energy extension, equality of closed smeared fields, and reconstruction of the original vacuum VOA. Formalizing the consequences of an explicitly supplied reconstruction interface would be useful. It would not be a formal proof of that theorem.

A1: pairwise annulus compatibility and positivity. The compatibility assertion is Proposition I.5 (p. 64); its domain and continuation steps are Lemmas I.6 and I.7. The final application is Theorem I.8. The revised proof deliberately constructs a lift for each compatible annulus pair. It does not require a global canonical choice of lifts for every annulus. A suitable formal interface would record the following data and laws:

1. the centrally extended annulus semigroup and its localized subsemigroups, together with their positive-energy representations;

2. the common localizer δ, localized lifts D_A, D_B, and the pair $\widetilde{A} = \widetilde{\delta}^{-1} D_A$, $\widetilde{B} = D_B \widetilde{\delta}$, so that $\widetilde{B}\widetilde{A} = D_B D_A$ is canonically localized;

3. the equality of local annulus actions with the represented vacuum operators, incoming insertions, sewing, and the outgoing unitary adjoint formula;

4. the round-annulus factorization giving both finite-energy range inclusion and density of the transported form domain;

5. holomorphy on the sewn interior, coverage by all allowed geometric representatives, and the no-insertion generators.

The deck ambiguity must be retained: changing the common lift conjugates the inserted field by $e^{2\pi i m L_0}$. Integer VOA weights make this conjugation trivial even when module lowest weights are fractional. Normalizing a vacuum operator alone does not eliminate the deck group on those modules. This small algebraic cancellation is a realistic formal target once the grading and representation laws are explicit.

The analytic continuation step can use `AnalyticOnNhd.eqOn_of_preconnected_of_eventuallyEq` from `Mathlib/Analysis/Analytic/Uniqueness.lean`, applied to scalar matrix coefficients after all holomorphy and component hypotheses are supplied. The finite-energy range and dense-domain lemmas can also be isolated from annulus geometry. The final equality of concrete nets is an elementary conditional commutant argument:

$$\mathcal{A}^{\mathrm{bl}}(I) \subset \mathcal{N}(I) \subset \mathcal{A}^{\mathrm{bl}}(I')' = \mathcal{A}^{\mathrm{bl}}(I).$$

It explicitly uses locality and Haag duality on the same vacuum Hilbert space; an abstract isomorphism of the two nets would not suffice.

These smaller arguments do not build the central extension, integrate Virasoro representations, prove geometric sewing, or prove positivity of Connes fusion and the resulting canonical VOA fusion pairings. Those are the large missing foundations. An honest intermediate formalization would name the exact integration, reconstruction, and sewing hypotheses, prove their pairwise consequence, and keep the positive-fusion theorem separate until it too is formalized.

1.5 Categories, extensions, and the independent Evans–Gannon proof

Mathlib has substantial general categorical infrastructure. Verified examples include `CategoryTheory.BraidedCategory` in `Mathlib/CategoryTheory/Monoidal/Braided/Basic.lean`, `CategoryTheory.RigidCategory` and duality interfaces in `Mathlib/CategoryTheory/Monoidal/Rigid/Basic.lean`, and `CategoryTheory.HalfBraiding`, `CategoryTheory.Center`, `CategoryTheory.Center.forget`, and `CategoryTheory.Center.ofBraided` in `Mathlib/CategoryTheory/Monoidal/Center.lean`. These permit genuine reuse in Appendices G, J, and K. They do not by themselves provide a unitary fusion category with finite simple-object data, positive spherical dimensions, modular S, T, a Q-system correspondence, or the Haagerup classification. No such complete manuscript-level framework was located in the local source audit.

A finite based ring can be formalized before its categorical realization. One can then prove the conditional dimension equations and obstruction calculations, keeping separate the theorem that a given unitary category realizes that ring. Similarly, transporting an already constructed algebra along a braided equivalence is distinct from proving that it yields a local net or a unitary VOA extension. The formal target for Appendix K must preserve the manuscript's scope: the original embedded extension and the chosen $\mathcal{H}_3$ extension need not be the same algebra; the 36 bimodule defects preserve U; and the holomorphic-net construction is not identified with the E_8 net by the VOA classification theorem.

The existing Evans–Gannon development should be credited precisely. In `lean/HaagerupIzumi/Analytic/GammaConstruction.lean`, `HaagerupIzumi.GammaContinuation.hypGamma` constructs the hyperbolic-gamma interface from the proved integral development. `HaagerupIzum`

`i.ruijsenaarsHypGamma` specializes it to the required periods, and `HaagerupIzumi.EGEquations_ruijsenaars` proves the equations for the resulting concrete coefficient data. The theorem `HaagerupIzumi.exists_EGEquations_odd` assumes only `Oddn` and produces complex coefficient data satisfying all the packaged Evans–Gannon equations. The source genuinely supplies the earlier analytic interface; this final theorem does not assume an unconstructed gamma function.

The packaged proposition `HaagerupIzumi.CoeffData.EGEquations` is defined in `lean/HaagerupIzumi/Completion.lean`; it records the linear symmetries and axis sums together with the quadratic, cubic, and quartic equations. The project README explicitly stops before the reconstruction theorem and categorical conclusions of the separate infinitude manuscript. Thus this development does not prove exhaustion of rank-six fusion categories, construct the present lattice subVOA, or establish its complete unitarity. Its finite Fourier infrastructure can be reused: the local interpolation proof already uses `ZMod.dft` and related declarations from `Mathlib/Analysis/Fourier/ZMod.lean`. Its contour machinery may be useful for appropriate scalar analytic sublemmas, but it is not an annulus or conformal-net construction.

1.6 A staged program with explicit completion criteria

The following stages give reviewable deliverables without suggesting that a short final Lean file would amount to a short foundational proof. Some stages can proceed independently; their applications must retain unproved inputs visibly.

1. **Certify finite data.** Define the exact coefficient algebra, sparse vectors, and matrix witnesses; prove a small checker's soundness; import and verify selected residual and rank certificates. The deliverable is a theorem about named finite data, with the interpretation map stated. It is not yet a theorem about the actual VOA.

2. **Complete the current-algebra layer.** Formalize the bounded weight argument for joint nilpotence, the singlet projection, the charged-module spanning deductions, and the top-table arithmetic. Connect these to the already checked saturation and inequality lemmas. The deliverable removes the corresponding finite algebraic assumptions.

3. **Formalize the recursive closure criterion.** Give a precise graded mode-algebra interface and prove the filtered Wick and Jacobi reductions. Instantiate the finite gates through the certified data. Construction of the genuine lattice VOA and validation of that instantiation remain explicit subsequent obligations.

4. **Separate representation theory from finite consequences.** Formalize the core-top and branching deductions, the based-ring classification, and character uniqueness under stated VOA and categorical interfaces. Keep optional categorical exhaustion as a separate parameter. The deliverable should make clear which deep representation-theory theorems remain inputs.

5. **Formalize the analytic lemmas.** Prove common-core reduction, adjoint commutation, dense-domain transport, grade-wise form comparison, deck cancellation, and the conditional Haag-duality squeeze. State reconstruction and annulus comparison with their full domains and normalizations. These supply reusable analysis without pretending to formalize the entire net/VOA correspondence.

6. **Discharge the foundational interfaces.** Build the required VOA, module tensor, conformal-net, annulus, and unitary fusion theories; formalize the cited reconstruction, classification,

positivity, and extension theorems; construct the actual objects and connect the finite certificates. Only completion of these dependencies supports an end-to-end Lean theorem matching the manuscript's full conclusion.

1.7 Assumptions, computation, and the trust boundary

A theorem with explicit parameters such as a spanning identity or a reconstruction theorem can be fully kernel-checked as an implication. Its parameters are not new global logical axioms. Nevertheless, the application to the CFT is unproved until those parameters are supplied. An interface whose fields simply repeat the desired conclusion adds little verification; useful interfaces expose the smaller mathematical laws from which the conclusion is actually derived.

An admitted `sorry` introduces `sorryAx`; a declared `axiom` installs an unproved proposition. Neither is equivalent to a theorem with all premises discharged. A source search of the local `HaagerupIzumi` Lean files found no uses of `sorry`, new `axiom` declarations, or `native_decide`; the saved axiom reports give the more relevant dependency information for the specific checked conclusions. A text search alone is not a substitute for an axiom audit of the final theorem.

There is a version-specific point about computation. In this Lean 4.33.1 installation, `native_decide` calls `Lean.Meta.nativeEqTrue`; the implementation in `Lean/Meta/Native.lean` compiles and evaluates a closed Boolean and, on success, introduces an auxiliary axiom asserting that its value is true. The kernel then checks a proof relative to that axiom; it does not independently replay the native computation by ordinary reduction. The older `Lean.ofReduceBool` mechanism is deprecated in this toolchain. Consequently a future certificate replay using `native_decide` would need an explicit expanded trust statement, even if it is useful operationally.

By contrast, ordinary proof-producing tactics and a `decide` proof checked by kernel reduction can verify an exact certificate without that native-evaluation axiom, subject to the final dependency report. For large data, the practical target is a small sound checker and manageable proof certificates, not necessarily replaying the original search algorithm inside the kernel. A Boolean printed by `#eval`, a Python exact calculation, a successful hash check, and a theorem with a kernel-checked proof are different artifacts. They should remain separately labeled in the ledger.

Future compiler and certificate runs must retain the repository's shared-desktop resource policy and run serially through its guarded launcher. The preserved baseline assessment introduced no new Lean declarations and does not claim any additional kernel-checked results beyond the recorded files above.

2 Additional assessment of the physics revision

The new physics sections provide much more explicit mathematical data: character coefficients, Virasoro-primary multiplicities, spin-resolved spectra, thermal intervals, low-weight couplings, rational correlators, fusion coefficients, defects, and boundary partition functions. None of the Python producers in `c8_physics_20260927/` is a Lean proof or has a Lean-verified implementation. Their exact arithmetic and internal checks improve reviewability and supply concrete inputs for formalization; they do not enlarge the scopes of the five existing CFT declarations listed in the preceding assessment. A separate new file adds the two small thermal-comparison lemmas stated below, bringing the selected CFT coverage to seven principal declarations in three files. The five baseline declarations retain their original scope.

2.1 Current manuscript anchors and historical concordance

The linked numbers, titles, and pages in the following table refer to the current manuscript. The first column is only a concordance for readers of the archived 55-page PDF; its former section numbers are not operative references in this guide. The current document places the mathematical development in appendices after the physics sections.

Archived label	Current manuscript title	Current anchor
1	Introduction and statement of results	Section 1, p. 4
2	The explicit lattice construction	Appendix A, p. 28
3	Finite closure and the all-weight criterion	Appendix B, p. 31
4	Gate 4: exact seeds, raising, and singlet recovery	Appendix C, p. 34
5	Gate 5 from Jacobi identities	Appendix D, p. 36
6	From the finite gates to properness	Appendix E, p. 38
7	Regularity from a positive conformal-vector identity	Appendix F, p. 39
8	Branching and the boundary fusion ring	Appendix G, p. 44
9	The complete character vector	Appendix H, p. 58
10	From vacuum unitarity to complete unitarity	Appendix I, p. 59
11	Unitary categorical identifications	Appendix J, p. 67
12	Full net theory and a chosen E_8 VOA extension	Appendix K, p. 69
13	Relation to the proposed Haagerup theories	Appendix L, p. 71
A	Certificate ledger and reproducibility	Appendix M, p. 71
B	Limited formalization	Appendix N, p. 74
C	Compilation and review scope	Appendix O, p. 75

The added physics material has the following current anchors. “Focused”, “substantial”, and “foundational” retain the dependency-based meanings stated earlier.

Current section title	Current anchor
Characters and a spin-resolved physical spectrum	Section 2, p. 7
Exact correlators and channel data	Section 3, p. 13
A complete weight-two chiral three-point dataset	Section 4, p. 17
Modular data, topological defects, and conformal boundaries	Section 5, p. 20
The Haagerup symmetry of the holomorphic extension	Section 6, p. 23
Perturbations, bootstrap coordinates, and use of the data	Section 7, p. 26

2.2 Characters, Virasoro primaries, and the full spectrum

Section 2, *Characters and a spin-resolved physical spectrum*, is in `c8_physics_20260927/spectrum.tex`. Its producer `spectrum.py` checks the published initial coefficients, the rational matrix recurrence, nonnegative integrality through the requested grade, descendant reconstruction, parity of the resulting spin spectrum, and an independent E_4/η^8 identity. The default data include 25 coefficients for each of twelve characters, complete full-CFT spectra through $\Delta = 24$, and an independent E_8 spectrum. These are finite arithmetic conclusions conditional on the input matrices and their character interpretation.

Physics argument	Missing Lean theorem and difficulty
Matrix recurrence	Define the twelve-dimensional exponent/characteristic data and a finite rational recurrence, prove nonzero denominators, and prove that its finite output agrees with the formal differential equation. This is focused to substantial. Identifying the resulting vector with the actual VOA trace functions still requires the foundational character theorem assessed under Appendix H.
Virasoro-primary subtraction	The finite Euler-product convolution and its inverse are focused. Their interpretation needs the irreducible $c = 8$ Virasoro characters, the Kac determinant or an equivalent irreducibility theorem, and the unitary highest-weight decomposition of each module. These are substantial to foundational representation-theoretic inputs.
Complete spin-resolved cutoff	Given nonnegative lowest weights and coefficient functions, prove the product formula and that no grade above N contributes to $\Delta \leq N$. The finite sums, parity identity, and relevant/marginal counts are focused. The tensor-product Hilbert-space interpretation and the diagonal modular invariant are additional premises.
E_8 comparison and invariant counts	A finite identity between the oscillator series and the selected character combination is focused. It does not identify the extension algebra. Counts $1488 = 6 \cdot 248$ for one chiral Haagerup action and $36 = 6^2$ for both independent actions additionally require the precise invariant-space theorem. That categorical interpretation belongs to the separate defect formalization below.

There is useful existing power-series infrastructure. `PowerSeries.coeff_mk` and `PowerSeries.coeff_mul` occur in `Mathlib/RingTheory/PowerSeries/Basic.lean`. The file `Mathlib/Combinatorics/Enumerative/Pentagonal/PowerSeries.lean` contains the Euler-product development, including `PowerSeries.coeff_prod_one_sub_X_pow_eventually_eq` and `PowerSeries.WithPiTopology.hasProd_one_sub_X_pow`. These declarations were read in the pinned checkout. They support formal coefficient manipulations; they do not state that these coefficients count this CFT's Virasoro primaries.

An economical formalization would first take an abstract sequence of graded dimensions and prove the descendant-removal and tensor-product counting lemmas. A second layer would connect that sequence to the Evans–Gannon recurrence. The final interpretation theorem must connect both layers to the actual modules. Merely importing the printed integers as definitions and proving that their sum is 2941 would check arithmetic but leave the substantial spectral assertion open.

2.3 Thermal bounds and their numerical intervals

Proposition 2.1 (p. 11) proves the modular heavy-tail estimate in Section 2. For $E = \Delta - 2/3$, an exact cutoff $D > 2/3$, and $b' = 4\pi^2/b$, it uses

$$Z = L_D + H_D, \qquad Z(b) = Z(b'), \qquad H_D(b) \leq e^{-(b-b')(D-2/3)} H_D(b')$$

to bound the tail at $b > 2\pi$. Two real-number steps are now checked in `lean/HaagerupIzumi/CFT/PhysicsBounds.lean`, in namespace `HaagerupIzumi.CFT`:

- `modular_heavy_tail_comparison` assumes $L + H = L_d + H_d$, $H \le rH_d$, and $0 \le r < 1$, and proves $H \le r(L - L_d)/(1 - r)$;
- `selfdual_tail_first_moment_comparison` assumes $E_{\mathrm{cut}} > 0$ and $E_{\mathrm{cut}} H \le L'$, and proves $H \le L'/E_{\mathrm{cut}}$.

The successful guarded compiler record is `c8_physics_20260927/physics_lean.log`, with sampled peak group RSS 1.549 GiB. Both axiom reports contain only `propext`, `Classical.choice`, and `Quot.sound`; the source contains neither `sorry` nor new axioms. The nonnegative contraction range is retained explicitly, although the first rearrangement only needs $r < 1$ once its comparison is supplied.

These lemmas do not derive the comparison from a spectral sum or identify r with the exponential in the application. The actual infinite-sum theorem remains more substantial: it needs convergence, termwise positivity, completeness of the finite cutoff, and the modular functional equation. Its self-dual form also needs differentiability and the identity $Z'(2\pi) = 0$. The first and second moment bounds require justified moment convergence and the elementary optimization of $E^k e^{-aE}$ on a positive half-line.

The Python output is a conditional interval certificate. Its positive spectrum and modular invariance are mathematical premises, not facts established by evaluating finitely many coefficients. The implementation uses directed decimal arithmetic, a fixed rational enclosure of π, and correctly rounded exponential/logarithm functions; the displayed six-decimal table is accepted only when every rigorous interval lies within 10^{-6} of its printed value. None of that implementation is currently kernel checked.

For a Lean version, a small rational interval checker with proved exponential and logarithm enclosures would be preferable to treating Python's decimal library as an axiom. One can separately formalize the analytic tail theorem over the real numbers and feed it certified finite moment bounds. Existing real analysis and infinite-sum infrastructure is relevant, but the complete pipeline from characters to the saved numerical intervals is new work. The two checked implications are the completed local algebraic part; sound interval evaluation and its analytic interpretation are substantial. The CFT assumptions still inherit the foundational requirements of the earlier sections.

2.4 Currents, lattice correlators, and low-weight couplings

The correlator and coupling material in Sections 3 and 4 is supplied by `c8_physics_20260927/correlators.tex` and `c8_physics_20260927/couplings.tex`. Its data distinguish direct finite lattice-mode products, rational identities reconstructed from saved singular OPEs, and coefficients completed using current Ward identities. Those three forms of evidence have different interpretation obligations; the final reports specify the supplied bases and direct probes.

In particular the completed weight-two dataset in `c8_physics_20260927/full_weight_two/` uses all 113 chiral Virasoro primaries. Its saved report records 1,442,897 ordered tensor slots, 18,365 nonzero entries, 51,076 current Ward checks, and 33 additional direct lattice-mode probes beyond the seed calculation. This supplies substantially more explicit input than a list of summed OPE coefficients. Those successful Python checks remain separate from the Lean obligations below.

The following finite targets are realistic once the coefficient algebra and basis conventions are represented in Lean:

- singular current brackets, levels, gauge factors, pairings, and the orthogonality and positivity of the stated finite Gram matrices;
- polynomial identities obtained by clearing denominators in the displayed rational four-point functions, with collision loci excluded when interpreting them as functions;
- exact extraction of three-point coefficients and their squared unit-norm values, retaining the specified invariant/Hermitian pairing and adjoint conventions;
- the finite Ward recursion, its compatibility with both current factors, and uniqueness of its completion from the supplied seed slice;
- the completeness of the chosen low-weight primary subspace, provided its exact rank and descendant-complement certificates.

These are substantial finite algebra rather than a need to formalize all CFT at once. The bridge from encoded lattice words to actual mode operators is the same missing bridge described for the finite gates. A successful check of a completed tensor against Ward identities establishes consistency of the completion. Its identification with all actual mode products also needs a proved determination theorem and authenticated seed products; a finite selection of direct probes does not by itself check every omitted product.

Beyond that finite layer, one needs the state-field correspondence, locality, conformal covariance, OPE convergence or formal expansion identities, and an explanation of how the chiral functions assemble into local bulk correlators. The known Potts and parafermion inputs also need their formal embeddings. These are foundational. A chiral vacuum correlator or a chosen lattice intertwiner block is not by itself the four-point function of the lightest diagonal bulk scalar. The formal statement must preserve the external fields and their normalization rather than silently promote one kind of block to the other.

No claim that a dimension-two scalar is exactly marginal follows from its existence or its three-point coefficient alone. A formal statement about a conformal manifold would additionally require a precise deformation theory, beta functions, and the appropriate all-orders argument. That subject is outside both the present Lean files and the finite arithmetic producer.

2.5 Modular data, defects, and conformal boundaries

Sections 5 and 6, *Modular data, topological defects, and conformal boundaries* and *The Haagerup symmetry of the holomorphic extension*, are in `c8_physics_20260927/defects.tex`. Their producer `defects.py` computes in $\mathbb{Q}(\zeta_{13})$, with exact Verlinde coefficients, modular relations, the forgetful multiplicities, and boundary/defect spectral tables. These checks establish finite algebraic properties of the input matrices and based rings, not the existence of the required associator or a conformal defect.

Physics argument	Missing Lean theorem and difficulty
Cyclotomic modular arithmetic	Prove that the twelve-coordinate quotient representation is a field embedded in $\mathbb{C}$, fix the primitive root and conjugation, and verify the Gauss-sum square root and sign. Then $S^2 = 1$, the twist relation, Verlinde integrality, and fusion associativity are substantial but bounded exact computations.

Physics argument	Missing Lean theorem and difficulty
Diagonal Verlinde lines	Given a unitary modular tensor category and the diagonal full CFT, derive line eigenvalues, twisted character matrices, and their sector multiplicities. The scalar sums are focused; the theorem that these matrices describe actual topological defects requires the categorical/full-CFT construction and is foundational.
Cardy boundaries and annuli	Formalize the boundary-state normalization, open/closed-channel identity, and the positive square roots entering g factors. Finite character multiplicities are focused. Their interpretation as boundary Hilbert spaces requires the Cardy/extension theorem. Formal verification of annulus integers is not that theorem.
Selected chiral $\mathcal{H}_3$ action	Prove the center-forgetful correspondence for the chosen extension, the bimodule/full-defect equivalence, and the identification of tube multiplicity spaces with $M_\mu \otimes \overline{H'}$. These are foundational categorical inputs. Once supplied, the 6×12 forgetful ring-map check and charge matrices are substantial finite algebra.
Invariant and charged gaps	Under the correct action, derive the fixed spaces and count their lowest scalar grades. This is focused after the sector assignment. It must distinguish one chiral action from both independent factors, and tube multiplicity spaces from their multiple copies in actual defect Hilbert spaces. Modular covariance alone is insufficient.
Six extension boundaries	Verify the finite annulus coefficients from the forgetful functor and the six-object fusion ring. The data-to-category interface must retain the actual module category, duals, and boundary normalization. The finite arithmetic is substantial; constructing these boundaries as part of the full CFT remains foundational.

The pinned checkout defines `CyclotomicField` and `IsCyclotomicExtension` in `Mathlib/NumberTheory/Cyclotomic/Basic.lean`; it also has primitive-root, complex-root-of-unity, and cyclotomic-polynomial developments. Thus the field-theory layer need not start from scratch. It still takes work to prove that the producer's concrete coordinate arithmetic and chosen complex embedding agree with those structures. An arbitrary root of the polynomial without the intended complex embedding does not determine the positive quantum dimensions used by the physics interpretation.

2.6 Bootstrap placement and observable scope

The physics reading guide in Section 7 distinguishes the diagonal full theory from the selected holomorphic extension and states which published bootstrap plots use which symmetry sectors. Formalizing the label correspondence, the central-charge convention $c_L = c_R = 8$, and a rational gap extracted from a proved spectrum would be focused. It would not formally verify a published numerical exclusion curve. That requires the original functional, its positivity on every required

conformal block, the truncation/error bounds, and a sound numerical or exact certificate. A plotted point read from a figure is evidence for comparison, not a kernel-checked exclusion or inclusion theorem.

The observable inventory is also an interface specification. Characters determine unflavoured multiplicities and, with analytic control, partition functions. They do not determine every normalized OPE coefficient or defect eigenvector. The new coupling and defect datasets close specific parts of that gap; each needs its own sound checker and interpretation theorem. Keeping those claims separate is essential to an eventual formal treatment that a physicist can use without reading the implementation.

2.7 A concrete order of formalization

A useful first deliverable is a small exact coefficient library covering the rational recurrence, finite Euler product, primary reconstruction, and cutoff-completeness lemma. Its theorem should expose the input character interpretation explicitly. A second deliverable is a sound finite algebra checker for the cyclotomic modular data and the supplied low-weight tensors, with an explicit interpretation map. A third is the real-analytic tail theorem and a certified interval evaluator. Each can be independently reviewed before the foundational VOA and defect interfaces are complete.

Completion criteria must be semantic: the formal theorem should refer to the actual character coefficients, mode products, or represented sectors through proved maps. File hashes, successful Python assertions, or a Lean theorem about unrelated constants cannot replace that connection. The two explicitly stated thermal-comparison lemmas are the additional Lean results of this physics revision; the much larger analytic and physical interpretations remain to be formalized.